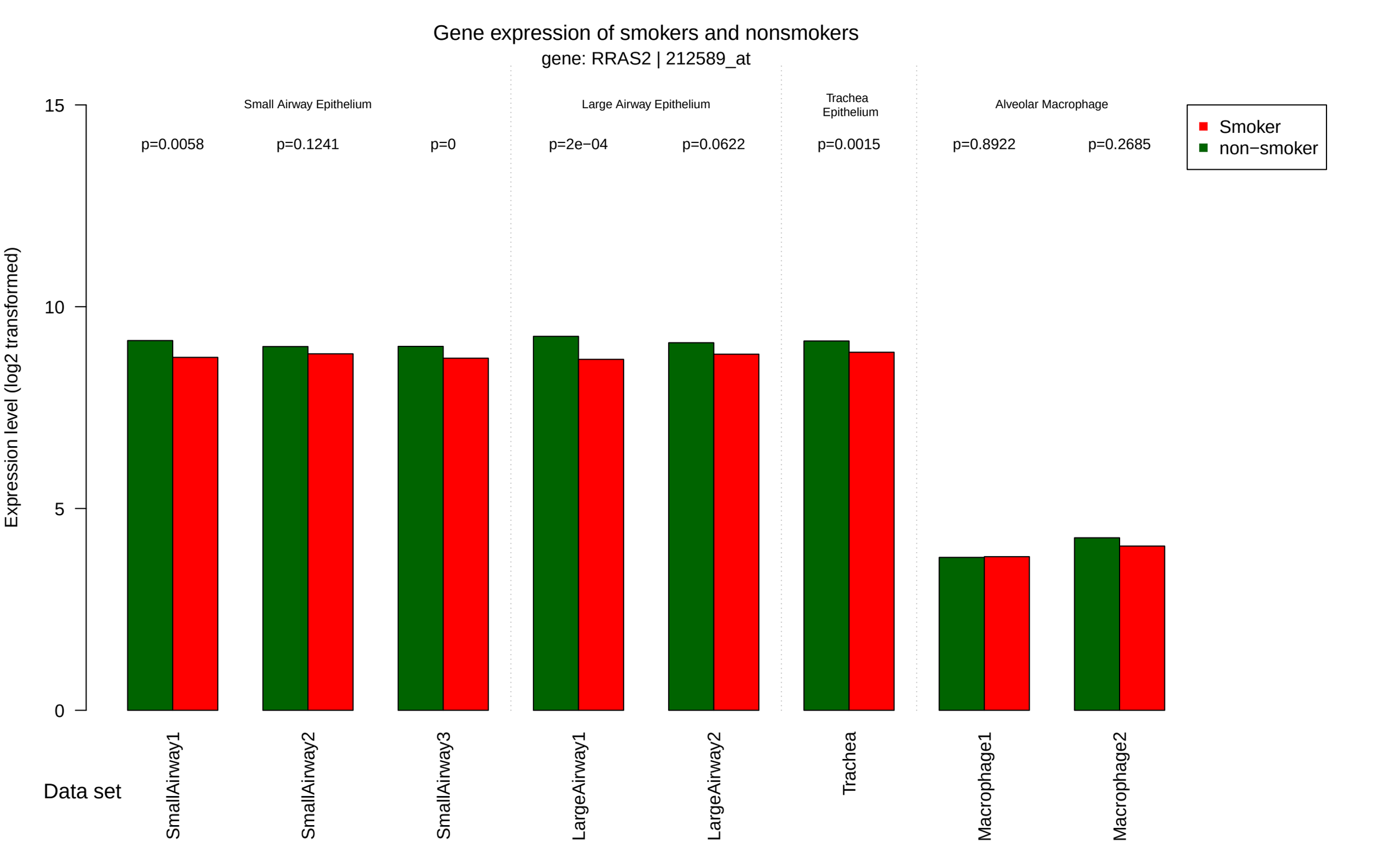
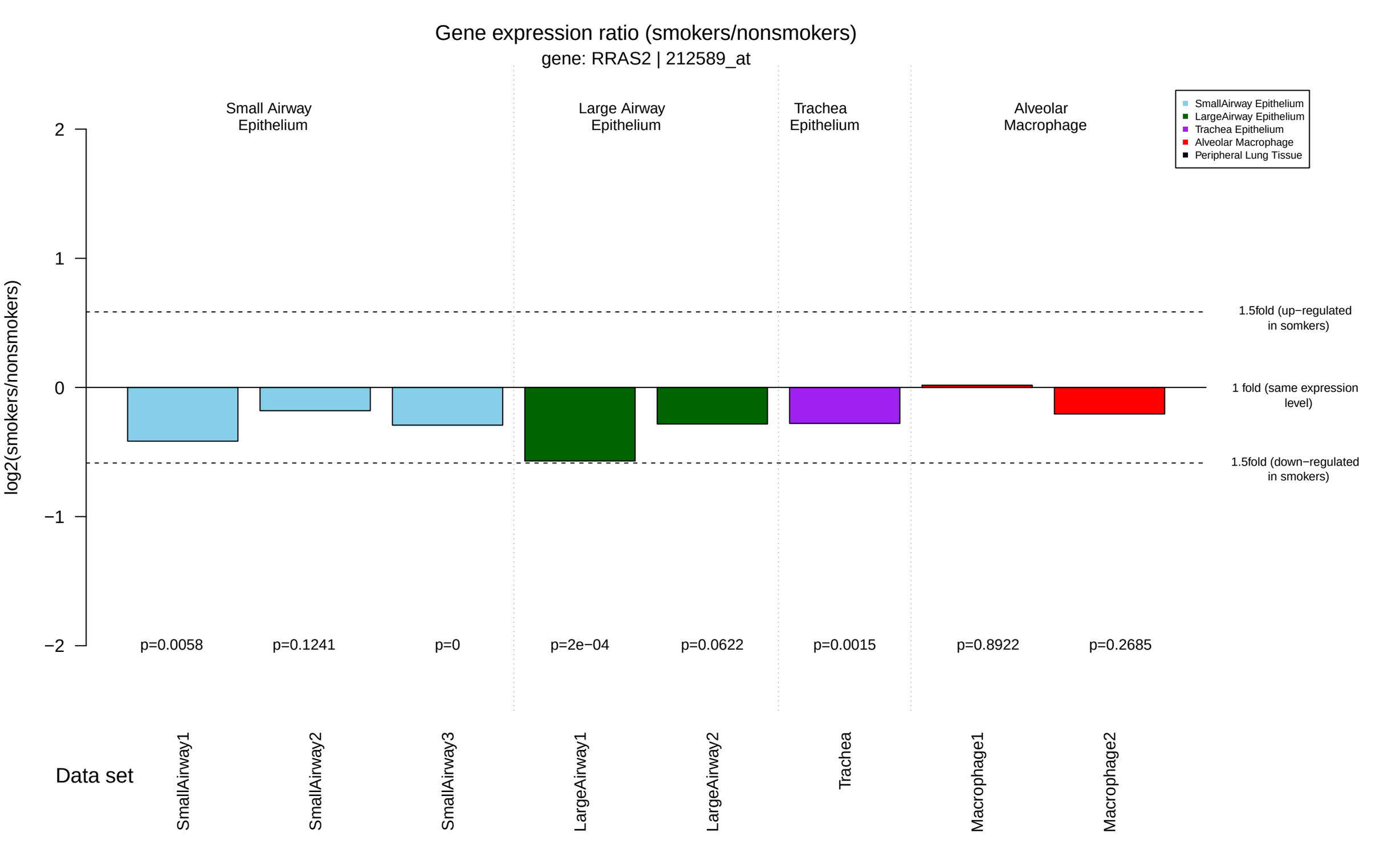
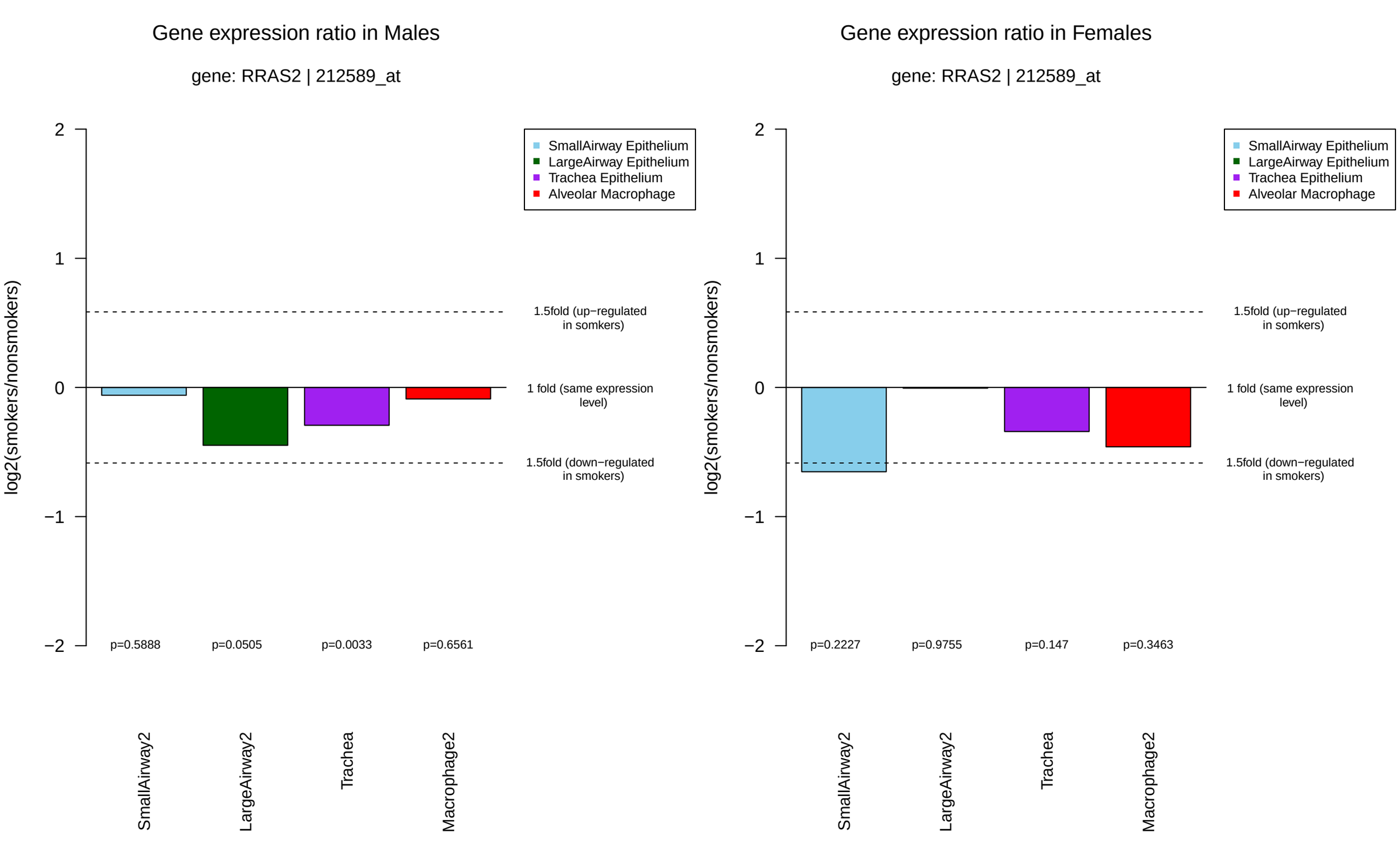
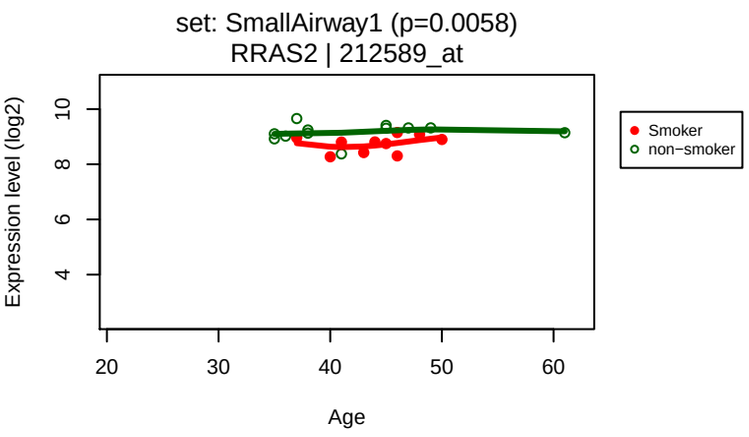
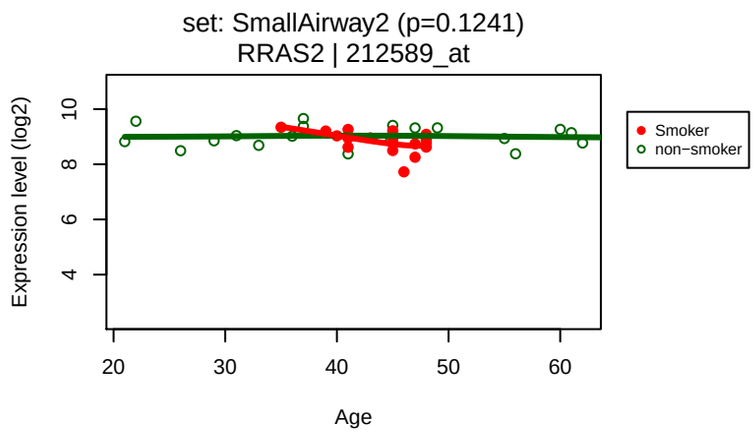
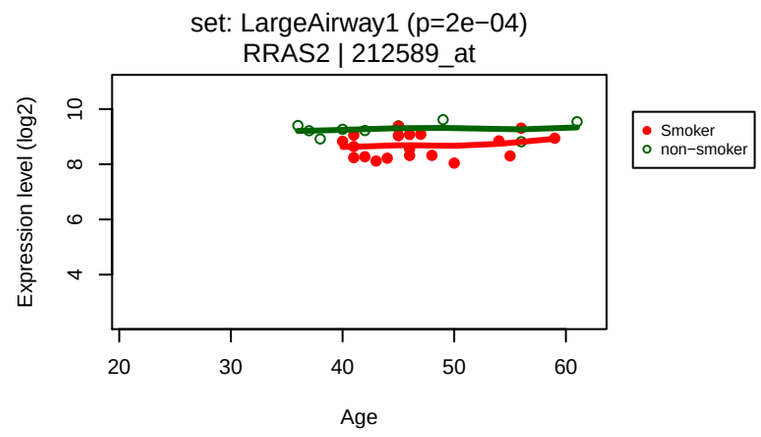

Gene expression of smokers and nonsmokers
gene: RRAS2 | 212589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0058
p=0.1241
p=0
p=2e−04
p=0.0622
p=0.0015
p=0.8922
p=0.2685
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

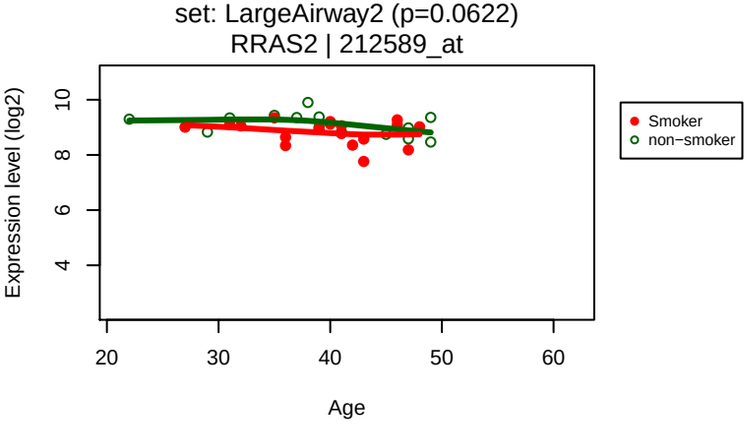

Gene expression ratio (smokers/nonsmokers)
gene: RRAS2 | 212589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0058
p=0.1241
p=0
p=2e−04
p=0.0622
p=0.0015
p=0.8922
p=0.2685
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

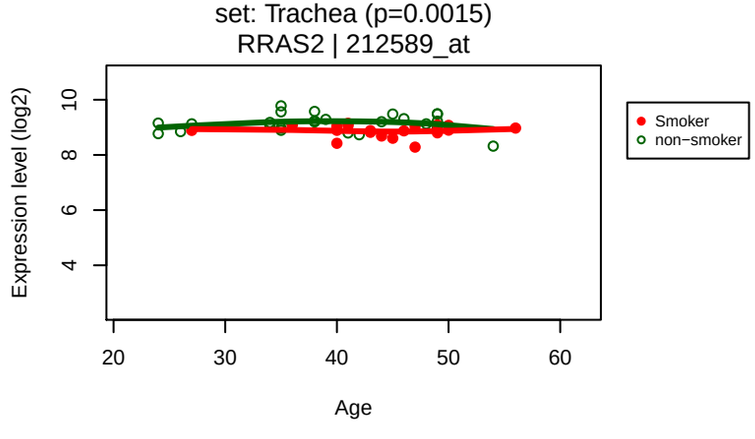

Gene expression ratio in Males
gene: RRAS2 | 212589_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5888
p=0.0505
p=0.0033
p=0.6561
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRAS2 | 212589_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2227
p=0.9755
p=0.147
p=0.3463
SmallAirway2
LargeAirway2
Trachea
Macrophage2

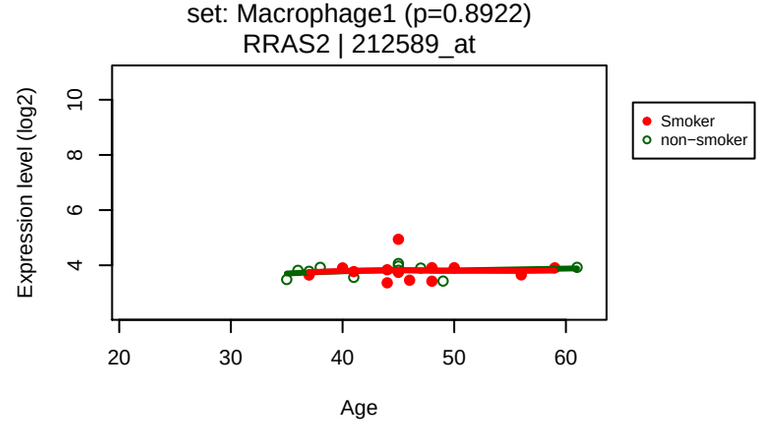

set: SmallAirway1 (p=0.0058)
RRAS2 | 212589_at
Expression level (log2)
Age
Smoker
non−smoker

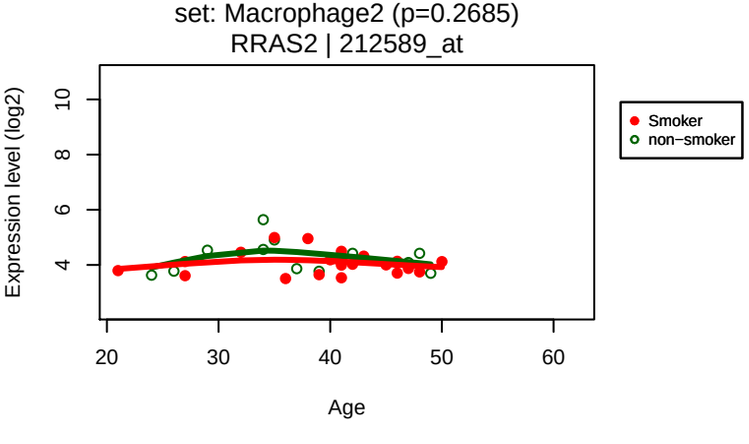

set: SmallAirway2 (p=0.1241)
RRAS2 | 212589_at
Expression level (log2)
Age
Smoker
non−smoker

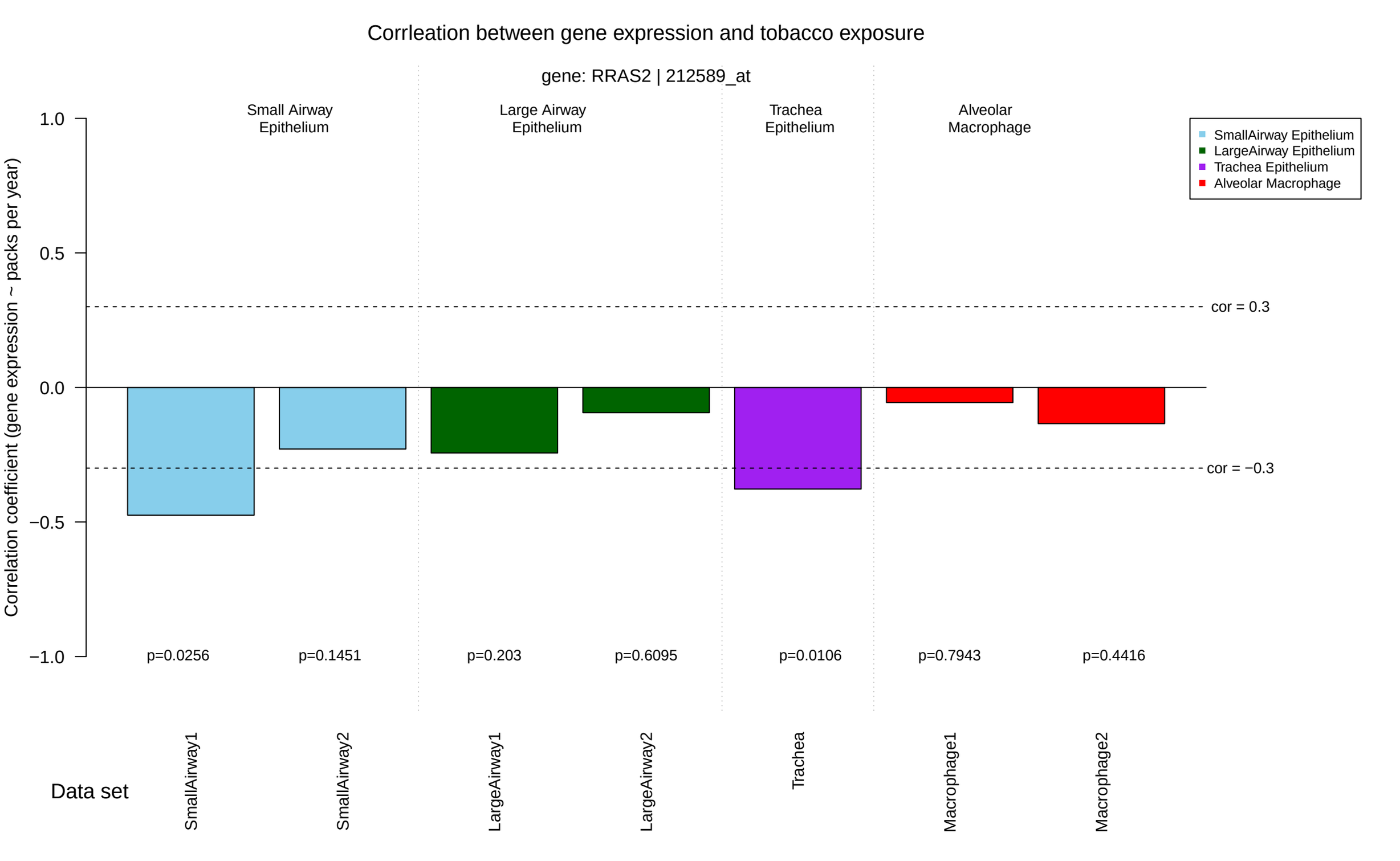

set: LargeAirway1 (p=2e−04)
RRAS2 | 212589_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0622)
RRAS2 | 212589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0015)
RRAS2 | 212589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8922)
RRAS2 | 212589_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2685)
RRAS2 | 212589_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRAS2 | 212589_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0256
p=0.1451
p=0.203
p=0.6095
p=0.0106
p=0.7943
p=0.4416
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2